Finite Fields

Introduction to Cryptography

Finite Fields

- In abstract algebra, a **finite field** or **Galois field** (so named in honor of Évariste Galois [1811 – 1832]) is a field that contains only finitely many elements
- Finite fields are important in number theory, algebraic geometry, Galois theory, **cryptography**, and **coding theory**
- The finite fields are completely known

Finite Fields

1. **Polynomial Ring $GF_p[x]$**
2. Galois Field GF_4
3. Galois Fields GF_8 and GF_{16}
4. Galois Field GF_{p^n}
5. Multiplicative Group of GF_{p^n}

Galois Field GF_p

- **Example**

  Cayley tables of GF_2

$\oplus$	0	1
0	0	1
1	1	0

$\otimes$	0	1
0	0	0
1	0	1

- **Example** Solve

$$3x + y + 4z + 1 = 0$$
$$6x + 5y + 3z + 6 = 0$$
$$x + 4y + 2z + 5 = 0 \quad \text{over } GF_7$$

- Solution

$$\begin{aligned} x + 5y + 6z + 5 &= 0 \\ y + 3z + 6 &= 0 \\ z + 1 &= 0 \end{aligned} \Rightarrow \begin{aligned} x + 5 &= 0 \\ y + 3 &= 0 \\ z + 1 &= 0 \end{aligned} \Rightarrow \begin{aligned} x &= 2 \\ y &= 4 \\ z &= 6 \end{aligned}$$

Polynomial Ring $GF_p[x]$

- **Polynomial ring** $F[x]$
  - The set of all polynomials whose coefficients are in the field F with usual additions and multiplications
- **Example** $(2x^3+3x^2+5x+1)(5x^2+3x+2) =$
  $3x^5+3x^3+5x^2+6x+2$ in $GF_7[x]$
- **Definition** The degree, denoted as $\deg(p(x))$, of $p(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0 \neq 0$ is the largest m such that $a_m \neq 0$
- **Definition** $\deg(0) = -\infty$

Polynomial Ring $GF_p[x]$

- **Definition** The leading term of $p(x)$, denoted as $\mathrm{LT}(p(x))$, is the monomial $a_m x^m$ with the highest degree in $p(x)$
- **Proposition** For any nonzero $p(x), q(x) \in F[x]$,
  1) $p(x)q(x) \neq 0$
  2) $\deg(p(x)q(x)) = \deg(p(x)) + \deg(q(x))$
- Proof $\mathrm{LT}(p(x)q(x)) = \mathrm{LT}(p(x))\mathrm{LT}(q(x)) \neq 0$ since F is a field
- Note Not necessarily true if $F[x]$ is replaced by $R[x]$

Polynomial Ring $GF_p[x]$

- **Theorem** (Division Algorithm) Given $a(x), b(x) \in F[x]$, there exist $q(x)$ [quotient] , $r(x)$ [remainder] $\in F[x]$ such that $a(x) = b(x)\,q(x) + r(x)$ where $r(x) = 0$ or $\deg(r(x)) < \deg(b(x))$
- Proof (Sketch)
  - $a(x) = 0$: Take $q(x) = r(x) = 0$
  - $a(x) \neq 0$, $\deg(a(x)) < \deg(b(x))$: Take $q(x) = 0$ and $r(x) = a(x)$
  - $a(x) \neq 0$, $\deg(a(x)) \geq \deg(b(x))$:
    Let $q_1(x) = \mathrm{LT}(a(x))/\mathrm{LT}(b(x)) \in F[x]$, then $a(x) = b(x)\,q_1(x) + a_1(x)$ where $a_1(x) = 0$ or $\deg(a_1(x)) < \deg(a(x))$.
    Continue until $a_k(x) = 0$ or $\deg(a_k(x)) < \deg(b(x))$.
    Then $r(x) = a_k(x)$ and $q(x) = q_1(x) + \ldots + q_k(x)$

Polynomial Ring $GF_p[x]$

- **Definition** $a(x), b(x), m(x) \in F[x]$, then $a(x)$ and $b(x)$ are congruent modulo $m(x)$ [denoted by $a(x) \equiv b(x) \pmod{m(x)}$] if $a(x) - b(x) \in \langle m(x) \rangle$
- **Example** Dividing $a(x) = 3x^4+5x+2$ by $b(x) = 2x^3+x^2+5$ in $GF_{11}[x]$, we get $a(x) = b(x)(7x+2) + (9x^2+3x+3)$, thus $a(x) \bmod b(x) = 9x^2+3x+3$ and $a(x) \equiv 9x^2+3x+3 \pmod{b(x)}$
- **Proposition** Euclidean Algorithm and Extended Euclidean Algorithm works in $F[x]$
  - i.e. given $a(x), b(x) \in F[x]$, there exist $u(x), v(x) \in F[x]$ such that $a(x)u(x) + b(x)v(x) = \gcd(a(x), b(x))$

Polynomial Ring $GF_p[x]$

- **Definition** An integral domain D is an Euclidean domain if for each nonzero $a \in D$ there exists a nonnegative $d(a) \in \boldsymbol{Z}$ such that
  1) If nonzero $a, b \in D$, then $d(a) \leq d(ab)$
  2) If $a, b \in D$ with $b \neq 0$, then there exist $q, r \in D$ such that $a = bq + r$, with $r = 0$ or $d(r) < d(b)$
- **Example**
  - $\boldsymbol{Z}$ with $d(a) = |a|$
  - $F[x]$ with $d(a(x)) = \deg(a(x))$
  - $\boldsymbol{Z}[i] = \{a + bi \mid a, b \in \boldsymbol{Z}\}$ with $d(a + bi) = |a + bi|^2 = a^2 + b^2$

Finite Fields

1. Polynomial Ring $GF_p[x]$
2. **Galois Field GF_4**
3. Galois Fields GF_8 and GF_{16}
4. Galois Field GF_{p^n}
5. Multiplicative Group of GF_{p^n}

Galois Field GF_4

- We will construct GF_4 in 3 ways
  1) Set of Linear Polynomials
  2) Set of Congruence Classes
  3) Linear Combinations of a Fixed Root
  - They look different, but essentially the same
- Note $\boldsymbol{Z}_4$ is *not* a field, since 2 does not have a multiplicative inverse

Galois Field GF_4

- **Definition** A non-constant polynomial $p(x) \in F[x]$ is irreducible if its only divisors are its associates $[c\,p(x)]$ and the nonzero constant polynomials
- $f(x) = x^2+x+1$ is irreducible over GF_2
  - It is confirmed by $f(0) = f(1) = 1 \neq 0$
    - Note that $f(x) = (x+2)^2$ over GF_3
    - Also $f(x) = (x-\omega)(x-\omega^2)$ over $\boldsymbol{C}$ where $\omega = (-1 \pm \sqrt{3}\, i)/2$
  - $p(x)$ mod $f(x) = 0, 1, x$, or $x+1$ for any $p(x) \in GF_2[x]$
- Q : (1) Is $f(x)$ irreducible over GF_{11} ?

  (2) For any $p(x) \in GF_{11}[x]$, $p(x)$ mod $f(x)$ = ?

Galois Field GF_4

1) GF_4 as a Set of Linear Polynomials

- $\{a_1x + a_2 \mid a_i = 0 \text{ or } 1\}$ with $\oplus$ and $\otimes$
- $(a_1x + a_2) \oplus (b_1x + b_2) = (a_1x + a_2) + (b_1x + b_2) \bmod 2$
- $(a_1x + a_2) \otimes (b_1x + b_2) = ((a_1x + a_2)(b_1x + b_2) \bmod x^2+x+1) \bmod 2$
- $f(x) = a_1x + a_2 \longleftrightarrow (a_1\,a_2)_2 \longleftrightarrow 2a_1+a_2$

$\otimes$	0	1	x	$x+1$
0	0	0	0	0
1	0	1	x	$x+1$
x	0	x	$x+1$	1
$x+1$	0	$x+1$	1	x

$\otimes$	00	01	10	11
00	00	00	00	00
01	00	01	10	11
10	00	10	11	01
11	00	11	01	10

$\otimes$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	3	1
3	0	3	1	2

- <u>Note</u> Every nonzero element has a multiplicative inverse

Galois Field GF_4

2) GF_4 as a Set of Congruence Classes

- For $f(x), g(x) \in GF_2[x]$, $f(x) \equiv g(x) \pmod{x^2+x+1}$ iff $f(x) - g(x) \in \langle x^2+x+1 \rangle$
- The congruence classes determined by the ideal $\langle x^2+x+1 \rangle$ are $[0]$, $[1]$, $[x]$, and $[x+1]$
- $[f(x)] \oplus [g(x)] = [f(x) + g(x)]$
- $[f(x)] \otimes [g(x)] = [f(x)\, g(x)]$
- $GF_4 \cong GF_2[x] / \langle x^2+x+1 \rangle$
  - Quotient ring

Galois Field GF_4

3) GF_4 as Linear Combinations of a Fixed Root

- Assume α is a root of x^2+x+1 over GF_2
  - $\alpha^2+\alpha+1=0$
  - Higher power of α can be reduced by $\alpha^2=\alpha+1$
    - e.g. $(\alpha+1)(\alpha+1)=\alpha^2+2\alpha+1=(\alpha+1)+1=\alpha$
- GF_4 is the set $\{0, 1, \alpha, \alpha+1\}$ with $+$ and $\times$ modulo 2, and the relation $\alpha^2=\alpha+1$

Finite Fields

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Galois Fields GF_8 and GF_{16}

- The elements of GF_{2^n} can be represented in a natural way as binary numbers easily understood by hardware
  - AES : GF_{2^8}
  - ECC : Elliptic Curves over GF_{2^n} or GF_p where p is a prime, not over GF_{p^n} where $p > 2$ and $n > 1$
- We look at GF_{2^n} for $n = 3$ and 4 now, then GF_{p^n} later
  - GF_8 can be represented as $a_2x^2+a_1x+a_0$, $a_i \in GF_2$
  - GF_{16} can be represented as $a_3x^3+a_2x^2+a_1x+a_0$, $a_i \in GF_2$
  - Additions are performed modulo 2
  - Multiplications are defined below

Galois Fields GF_8 and GF_{16}

1) $GF_8 = GF_{2^3}$

- x^3+x^2+1 and x^3+x+1 are the only irreducible polynomials over GF_2 of degree 3
  - neither 0 nor 1 is is a root of either polynomial
- $x^8 - x \equiv x(x+1)(x^3+x+1)(x^3+x^2+1) \pmod 2$ gives a complete list of all the irreducible polynomials of degree 1 and 3 over GF_2
- $GF_8 \cong GF_2[x] / \langle x^3+x^2+1 \rangle \cong GF_2[x] / \langle x^3+x+1 \rangle$
  - e.g. in $GF_2[x] / \langle x^3+x+1 \rangle$
    - $[x^2+1] \oplus [x^2+x] = [(x^2+1) + (x^2+x)] = [x+1]$
    - $[x^2+1] \otimes [x^2+x] = [(x^2+1)(x^2+x)] = [x^4+x^3+x^2+x]$
      $= [(x+1)(x^3+x+1) + (x+1)] = [x+1]$

Galois Fields GF_8 and GF_{16}

- Assume α is a root of x^3+x+1 over GF_2
  - $\alpha^3+\alpha+1 = 0$, hence $\alpha^3 = -\alpha-1 = \alpha+1 \pmod 2$
  - $\alpha^4 = \alpha(\alpha+1) = \alpha^2+\alpha$
  - α^5, α^6, and $\alpha^7 = 1$ are reduced similarly
  - $GF_8^* = \langle\alpha\rangle$
- Each field element is a root of $x^8 - x$ since

1) $0^8 - 0 = 0$
2) $(\alpha^i)^8 - (\alpha^i) = (\alpha^7)^i \times \alpha^i - \alpha^i = \alpha^i - \alpha^i = 0$

- α, α^2, and $\alpha^2+\alpha$ are roots of x^3+x+1
- $\alpha+1$, α^2+1, and $\alpha^2+\alpha+1$ are roots of x^3+x^2+1

Galois Fields GF_8 and GF_{16}

- **Definition** F is a subfield of K and $u \in K$, then the uniquely determined monic polynomial $g \in F[x]$ generating the ideal $J = \{ f \in F[x] \mid f(u) = 0 \}$ of $F[x]$ is the minimal polynomial of u over F
- **Example** $x^3 + x^2 + 1$ is the minimal polynomial of $\alpha^2 + 1$ over GF_2 where $\alpha^3 + \alpha + 1 = 0$, since
  - $x^3 + x^2 + 1$ is irreducible over GF_2
  - $(\alpha^2+1)^3 + (\alpha^2+1)^2 + 1 = \alpha^6 + 4\alpha^4 + 5\alpha^2 + 3$

    $\equiv (\alpha+1)^2 + \alpha^2 + 1 \equiv 0 \pmod 2$

Galois Fields GF_8 and GF_{16}

- Elements of GF_8

Power Notation	Polynomial Notation	Binary Notation	Decimal / Hex Notation	Minimal Polynomial
0	0	000	0	x
α^0	1	001	1	$x+1$
α^1	α	010	2	x^3+x+1
α^2	α^2	100	4	x^3+x+1
α^3	$\alpha+1$	011	3	x^3+x^2+1
α^4	$\alpha^2+\alpha$	110	6	x^3+x+1
α^5	$\alpha^2+\alpha+1$	111	7	x^3+x^2+1
α^6	α^2+1	101	5	x^3+x^2+1

Galois Fields GF_8 and GF_{16}

2) $GF_{16} = GF_{2^4}$

- x^4+x+1, x^4+x^3+1, and $x^4+x^3+x^2+x+1$ are the only irreducible polynomials over GF_2 of degree 4
- $x^{16} - x \equiv x(x+1)(x^2+x+1)(x^4+x+1)(x^4+x^3+1)(x^4+x^3+x^2+x+1)$ (mod 2) gives a complete list of all the irreducible polynomials of degree dividing 4, i.e., 1, 2, or 4
- $GF_{16} \cong GF_2[x] / \langle x^4+x+1 \rangle \cong GF_2[x] / \langle x^4+x^3+1 \rangle$ $\cong GF_2[x] / \langle x^4+x^3+x^2+x+1 \rangle$

Galois Fields GF_8 and GF_{16}

- Assume α is a root of x^4+x+1 over GF_2
  - $\alpha^4+\alpha+1=0$, hence $\alpha^4=-\alpha-1=\alpha+1 \pmod 2$
  - $GF_{16}^* = \langle\alpha\rangle$
- **Note** If β is a root of $x^4+x^3+x^2+x+1$, then $GF_{16}^* \neq \langle\beta\rangle$ since $\beta^5=1$
- **Fact**
  - $(a+b)^2 = a^2+2ab+b^2 = a^2+b^2$, $(a+b)^{2^r} = a^{2^r}+b^{2^r}$ in GF_{2^n}
  - If u is a root of $f(x) = c_m x^m + \ldots + c_1 x + c_0 \in GF_2[x]$, then u^{2^r} is also a root of $f(x)$ for $r \in \boldsymbol{N}$, since $f(u^{2^r}) = c_m(u^m)^{2^r} + \ldots + c_1 u^{2^r} + c_0$
    $= c_m^2(u^m)^{2^r} + \ldots + c_1^2 u^{2^r} + c_0^2 = (c_m(u^m)^{2^{r-1}} + \ldots + c_1 u^{2^{r-1}} + c_0)^2$
    $= \ldots = (c_m u^m + \ldots + c_1 u + c_0)^{2^r} = f(u)^{2^r} = 0$

Galois Fields GF_8 and GF_{16}

- Irreducible Polynomials for the Modulus 2

$n = 1$	e
10	1
11	1

$n = 2$	e
111	3

$n = 3$	e
1011	7
1101	7

$n = 4$	e
10011	15
11001	15
11111	5

$n = 5$	e
100101	31
101001	31
101111	31
110111	31
111011	31
111101	31

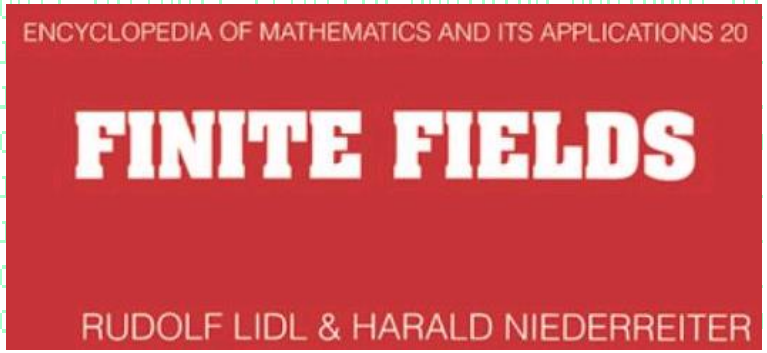
$n = 6$	e
1000011	63
1001001	9
1010111	21
1011011	63
1100001	63
1100111	63
1101101	63
1110011	63
1110101	21

$n = 7$	e
10000011	127
10001001	127
10001111	127
10010001	127
10011101	127
10100111	127
10101011	127
10111001	127
10111111	127
11000001	127
11001011	127
11010011	127
11010101	127
11100101	127
11101111	127
11110001	127
11110111	127
11111101	127

$n = 8$	e
100011011	51
100011101	255
100101011	255
100101101	255
100111001	17
100111111	85
101001101	255
101011111	255
101100011	255
101100101	255
101101001	255
101110001	255
101110111	85
101111011	85
110000111	255
110001011	85
110001101	255
110011111	51
110100011	85
110101001	255
110110001	51
110111101	85
111000011	255
111001111	255
111010111	17
111011101	85
111100111	255
111110011	51
111110101	255
111111001	85

Galois Fields GF_8 and GF_{16}

- **Definition** For nonzero $f \in GF_q[x]$ with $f(0) \neq 0$, the least positive integer e for which $f(x) \mid (x^e - 1)$ is the order of f and denoted by $\mathrm{ord}(f) = \mathrm{ord}(f(x))$
- **Theorem** f is irreducible of degree m and $f(0) \neq 0$, then $\mathrm{ord}(f)$ = the order of any root of f in the multiplicative group $GF_{q^m}{}^*$
- Proof See



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Galois Field GF_{p^n}

- **Definition** F is a subfield of K, $f(x) \in F[x]$ is of degree n, then K is the splitting field of $f(x)$ if K is the smallest field containing all roots of $f(x)$, i.e., $f(x) = a(x-\alpha_1)(x-\alpha_2)\ldots(x-\alpha_n)$ where $\alpha_i \in K$ for all i
- We will show that GF_{p^n} is the splitting field of $x^{p^n} - x$, which contains exactly p^n elements

Galois Field GF_{p^n}

- **Theorem**
  - Assumptions :
    1. p is a prime
    2. $q(x) \in GF_p[x]$ is irreducible with $\deg(q(x)) = n > 1$
    3. S is a set of all polynomials of the form
       $$a_{n-1}x^{n-1} + \ldots + a_1 x + a_0 \text{ where } a_i \in GF_p$$
    4. $f(x) \oplus g(x) = f(x) + g(x) \bmod p$
    5. $f(x) \otimes g(x) = (f(x)g(x) \bmod q(x)) \bmod p$
  - Conclusion : S with $\oplus$ and $\otimes$ is a field of p^n elements, which is GF_{p^n}

Galois Field GF_{p^n}

- Proof (Sketch)
  ① There are exactly p^n polynomials

  $$r(x) = a_{n-1}x^{n-1} + \ldots + a_1 x + a_0 \quad \text{where } a_i \in GF_p$$

  - These are the elements of GF_{p^n}

  ② For each nonzero $r(x) \in GF_p[x]$, there exists $s(x) \in GF_p[x]$ such that

  $$(r(x)s(x) \bmod q(x)) \bmod p = 1$$

  since $q(x)$ is irreducible

  - $s(x)$ is the multiplicative inverse of $r(x)$ in GF_{p^n}

Galois Field GF_{p^n}

- **Example** Which quotient is isomorphic to GF_{64} ?

A: $GF_2[x] / \langle x^6 + x^2 + 1 \rangle$ B: $GF_2[x] / \langle x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 \rangle$

C: $GF_2[x] / \langle x^6 + x^3 + 1 \rangle$ D: $GF_2[x] / \langle x^6 + x^4 + x^3 + x^2 + 1 \rangle$

Solution In $GF_2[x]$:

- $x^6 + x^2 + 1 = (x^3 + x + 1)^2$
- $x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = (x^3 + x^2 + 1)(x^3 + x + 1)$
- $x^6 + x^3 + 1$ is irreducible
- $x^6 + x^4 + x^3 + x^2 + 1 = (x^2 + x + 1)(x^4 + x^3 + x^2 + x + 1)$

Hence $GF_2[x] / \langle x^6 + x^3 + 1 \rangle$ is a field of order $2^6 = 64$

Galois Field GF_{p^n}

- **Corollary** If $f(x) \in GF_p[x]$, then GF_{p^n} contains a root of $f(x)$ for some n
- Proof (Sketch)
  - There exists an irreducible $q(x) \mid f(x)$
  - Use $q(x)$ to construct GF_{p^n} where $n = \deg(q(x))$
  - For $x \in GF_{p^n}$, $(f(x) \bmod q(x)) \bmod p = 0$ since $q(x) \mid f(x)$, hence x is a root of $f(x)$

Galois Field GF_{p^n}

- **Proposition** The set K consisting of the roots of $x^{p^n} - x$ with $+$ and $\times$ over GF_p is a field
- Proof
  - $0, 1 \in K$
  - Suppose α_1, α_2 are roots of $x^{p^n} - x$, then $\alpha_1^{p^n} = \alpha_1$, $\alpha_2^{p^n} = \alpha_2$
    - $\alpha_1 + \alpha_2 \in K$ since $(\alpha_1 + \alpha_2)^{p^n} = \alpha_1^{p^n} + \alpha_2^{p^n} = \alpha_1 + \alpha_2$
    - $\alpha_1 \alpha_2 \in K$ since $(\alpha_1 \alpha_2)^{p^n} = \alpha_1^{p^n} \alpha_2^{p^n} = \alpha_1 \alpha_2$
  - Suppose $\alpha \neq 0$ with $\alpha^{p^n} - \alpha = 0$
    - $-\alpha \in K$ since $(-\alpha)^{p^n} = \alpha^{p^n} = \alpha = -\alpha$ for $p = 2$, or
      $(-\alpha)^{p^n} = -\alpha^{p^n} = -\alpha$ for odd prime p
    - $\alpha^{-1} \in K$ since $(\alpha^{-1})^{p^n} = (\alpha^{p^n})^{-1} = \alpha^{-1}$

Galois Field GF_{p^n}

- **Lemma** $f(x)$ of degree n splitting completely has n distinct roots iff $f(x)$ and its (formal) derivative $f'(x)$ are relatively prime
- Proof Write $f(x) = \prod (x-\alpha_i)^{k_i}$ where α_i are distinct, then $f'(x) = \sum \left(k_i (x-\alpha_i)^{k_i-1} \prod_{j\neq i} (x-\alpha_j)^{k_j} \right)$
  - ($\Rightarrow$) $k_i = 1$ for all i, then $f'(\alpha_i) \neq 0$ for all i, so $f(x)$ and $f'(x)$ have no common factor
  - ($\Leftarrow$) $k_i > 1$ for some i, then $x-\alpha_i$ is a common factor

Galois Field GF_{p^n}

- **Remark** $f(x) = x^{p^n} - x$ has no multiple roots over GF_p since $f'(x) = p^n x^{p^n - 1} - 1 \equiv -1 \pmod{p}$ relatively prime to $f(x)$
- **Remark** The splitting field of $x^{p^n} - x$ over GF_p consists precisely of its p^n distinct roots
  - This is another representation of GF_{p^n}
- **Proposition** $f(x) \in GF_p[x]$ of degree n has at most n distinct roots in GF_{p^n}
- Proof Use the Division Algorithm

Galois Field GF_{p^n}

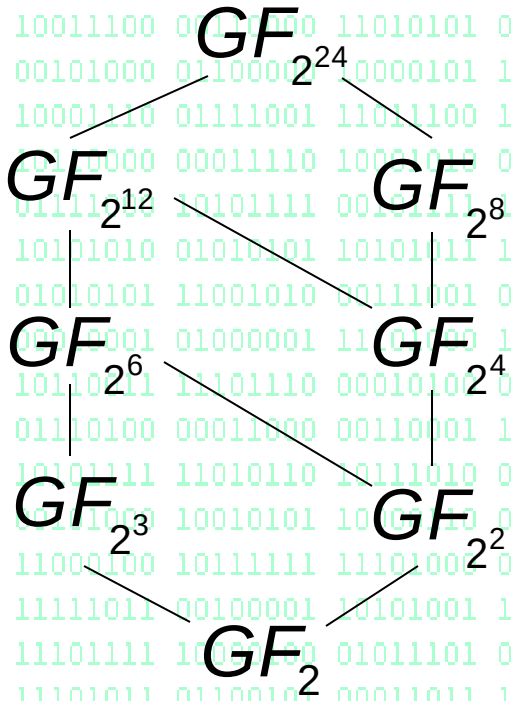
- **Remark** Let F be a field. Then any polynomial $f(x) \in F[x]$ has at most n roots
- Proof Use the Division Algorithm
- **Example**
  - In $\boldsymbol{Z}_8$, 1, 3, 5, 7 are roots of $x^2 - 1 = 0$
  - Note that $\boldsymbol{Z}_8$ is not a field

Galois Field GF_{p^n}

- **Theorem** For each prime p and positive integer n, there is exactly one field GF_{p^n} with p^n elements up to isomorphism, which is the splitting field of $x^{p^n} - x$ over GF_p
  - up to isomorphism : any two fields with p^n elements are isomorphic
- **Remark** GF_{p^n} is a vector space over GF_p with the basis $\{x^{n-1}, x^{n-2}, \ldots, x, 1\}$

Galois Field GF_{p^n}

- **Proposition** Any subfield of GF_{p^n} is of the form GF_{p^d}, $d|n$
- **Example**



Finite Fields

1. Polynomial Ring $GF_p[x]$
2. Galois Field GF_4
3. Galois Fields GF_8 and GF_{16}
4. Galois Field GF_{p^n}
5. **Multiplicative Group of GF_{p^n}**

Multiplicative Group of GF_{p^n}

- **Definition** G is a group, $a \in G$, the order of a [denoted as $o(a)$] is the smallest positive integer t such that $a^t = 1$
- **Note** If G is finite, then $o(a) = o(\langle a \rangle) \mid o(G)$ by Lagrange's Theorem
- **Theorem** $GF_{p^n}^* = \langle g \rangle$ for some $g \in GF_{p^n}$
  - We will prove the theorem in the following way:
    - For each $d \mid (p^n - 1)$, there are exactly $\phi(d)$ elements of order d in $GF_{p^n}^*$
    - In particular, there are $\phi(p^n - 1)$ generators

Multiplicative Group of GF_{p^n}

- Proof of Theorem (Sketch)
  - Denote $n(d)$ the number of elements in GF_{p^n}* of order d
  - For $d \mid (p^n - 1)$, we have $(x^d - 1) \mid (x^{p^n - 1} - 1)$
  - $n(q) = q - 1$ for prime q
  - $n(q^m) = q^{m-1}(q - 1)$
  - $n(ab) = n(a)\, n(b)$ for relatively prime a, b
  - $n(d) = \phi(d)$

Multiplicative Group of GF_{p^n}

- **Example** Since $P(x) = x^5 + 2x + 1$ is irreducible over GF_3, the quotient ring $K = GF_3[x]/(P(x))$ is a finite field
  - Let $Q(x) = x^2 + 2x + 1$
  - The number of elements in K is $|K| = 3^5 = 243$
  - $(K - \{0\}, \times)$ is a cyclic group of order 242
  - $Q(x)^{1213} = Q(x)^{242 \cdot 5 + 3} = Q(x)^3 = 2x^3 + x^2 + 2x + 1$ in K
  - Since $Q(x)(x^4 + x^3 + 2x + 1) - (x^5 + 2x + 1) \cdot x = 1$, we have $Q(x)^{-1} = x^4 + x^3 + 2x + 1$ in K

Multiplicative Group of GF_{p^n}

- **Definition** $f(x) \in GF_p[x]$ with $\deg(f(x)) = n$ is a primitive polynomial if it is the minimal polynomial of a primitive element [a generator of GF_{p^n}] in GF_{p^n}
- **Lemma** If f is the minimal polynomial of $\alpha \in GF_{p^n}$, then f is irreducible
- Proof
  - Suppose $f(x) = a(x)\,b(x)$ for some $a(x), b(x) \in GF_p[x]$
  - $f(\alpha) = 0 = a(\alpha)\,b(\alpha)$ implies $a(\alpha) = 0$ or $b(\alpha) = 0$
  - If $a(\alpha) = 0$, then $\deg(a(x)) = \deg(f(x))$ since $f(x)$ is the minimal polynomial of α, thus $b(x)$ is a constant
  - Similarly, if $b(\alpha) = 0$, then $a(x)$ is a constant
  - Thus $f(x)$ is irreducible

Multiplicative Group of GF_{p^n}

- **Proposition** The polynomial $x^{p^n} - x$ is precisely the product of all the distinct irreducible polynomials in $GF_p[x]$ of degree d where d runs through all divisors of n
- **Example**
  - $x^8 - x = x(x+1)(x^3+x+1)(x^3+x^2+1) \pmod 2$
  - $x^{16} - x \equiv x(x+1)(x^2+x+1)(x^4+x+1)(x^4+x^3+1)(x^4+x^3+x^2+x+1) \pmod 2$
  - Q: How many irreducible polynomials of degree 7 over GF_2?

Multiplicative Group of GF_{p^n}

- **Theorem** Given any prime p and $n \in P$, there exists at least one monic irreducible $f(x) \in GF_p[x]$ of degree n
- Proof (Sketch)
  - Let ω be a primitive element of GF_{p^n}, and $f(x)$ with $\deg(f(x)) = m$ be its minimal polynomial
  - $f(x)$ is irreducible by Lemma
  - We obtain $m = n$ by showing $m \geq n$ and $m \leq n$

Multiplicative Group of GF_{p^n}

- **Example**
  - To construct GF_{25}, we can use $g(x) = x^2+3 \in GF_5[x]$
    - $g(0), g(1), g(2), g(3), g(4) \bmod 5 = 3, 4, 2, 2, 4$ respectively
    - None of these are zero, hence x^2+3 is irreducible over GF_5
  - To get a primitive element, we should find $q(x)$ such that $(q(x)^{24} \bmod x^2+3) \bmod 5 = 1$ and $\neq 1$ with less power
    - Take $q(x) = x+3$
    - $((x+3)^8 \bmod x^2+3) \bmod 5 = 2x+2$
    - $((x+3)^{12} \bmod x^2+3) \bmod 5 = 4$
    - $((x+3)^{24} \bmod x^2+3) \bmod 5 = 1$
  - Q: Is $g(x)$ a primitive polynomial? If not, how to get one?